Weekly Putnam

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PROBLEM

Let n be a positive integer, and let M be the $n \times n$ matrix whose (i, j) entry, for $1 \le i, j \le n$, is 1 if the binomial coefficient $\binom{2i+j}{i}$ is odd and 0 if it is even. Let b be the number of blocks of consecutive 1's in the binary expansion of n (so $b = 2$ for $n = 11 = 1011_2$). Prove that

$$\det M = (-1)^{\lceil n/2 \rceil + b}.$$

Answer: $\det M = (-1)^{\lceil n/2 \rceil + b}$; equivalently $\det M = \prod_{k=1}^{n} \varepsilon_k$ with $\varepsilon_k = -1$ when the odd part of k is 3 mod 4 and $\varepsilon_k = 1$ otherwise (the paperfolding sequence). For $n = 2026$ the value is $+1$, for $n = 2027$ it is -1. · **Level:** A6/B6 · **Topic:** combinatorial linear algebra, binary digits

Hints

1. When is $\binom{2i+j}{i}$ odd? Lucas's theorem, or Kummer's carry criterion, turns the parity into a statement about the binary digits of i and of $i + j$.
2. Separate the rows, and the columns, according to the parity of their indices. To an even-indexed row, two columns whose indices differ by one look identical. Subtract one from the other.
3. The matrix left for the odd-indexed rows is not M again: it has an extra all-ones row on top. Split it in the same way, and watch the one column that both halves share; a block triangular shape appears, and the sign of the interleaving permutation is what you are after.

The key idea

> Lucas's theorem turns the entry into the indicator that i and $i + j$ have no binary digit 1 in common. Splitting rows and columns by the parity of their index reproduces the same kind of matrix at half the size, but only after an all-ones row has been adjoined, so the induction has to carry two companion matrices along. The shared column in the companion matrix forces a block triangular shape whose diagonal blocks are two copies of one matrix, so the determinant squares to 1, and all the information sits in the signs of two interleaving permutations. Summed over the binary levels of n, those signs give $\lceil n/2 \rceil + b$.

Solution

Throughout, $a \,\&\, c$ denotes the bitwise AND of nonnegative integers a and c, so $a \,\&\, c = 0$ means that a and c have no binary digit 1 in a common position. For integers $a, c \ge 0$ put

$$G(a, c) = \begin{cases} 1 & \text{if } a \,\&\, (a + c) = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Note that $G(0, c) = 1$ for every c, and $G(a, 0) = 1$ only for $a = 0$.

Lemma (Parity of the entries). *For $i, j \ge 1$, $\binom{2i+j}{i}$ is odd if and only if $G(i, j) = 1$. Hence $M = \big(G(i, j)\big)_{1 \le i, j \le n}$.*

Proof. Write $m = 2i + j = i + (i + j)$. By Lucas's theorem, $\binom{m}{i}$ is odd exactly when every binary digit 1 of i is also a digit 1 of m. We claim that for $a, c \ge 0$ this holds for $m = a + c$ exactly when $a \,\&\, c = 0$.

If $a \,\&\, c = 0$, then $a + c$ is the bitwise OR of a and c, which contains all digits of a. If $a \,\&\, c \neq 0$, let t be the lowest position where both a and c have a 1. Below position t the two numbers have no common 1, so adding them produces no carry into position t; at position t the two digits 1 add to 0 with a carry. So $a + c$ has digit 0 at position t while a has digit 1 there, and the claim follows. Apply it with $a = i$, $c = i + j$. □

We now use three families of matrices, all built from G:

$$M_n = \big(G(a,c)\big)_{1\le a\le n,\ 1\le c\le n}, \qquad R_k = \big(G(a,c)\big)_{0\le a\le k-1,\ 1\le c\le k}, \qquad P_k = \big(G(a,c)\big)_{0\le a\le k-1,\ 2\le c\le k+1},$$

with M_0, R_0, P_0 the empty matrix of determinant 1. So $M = M_n$; the matrix R_k is the same window shifted one row up, which gives it the all-ones row $a = 0$; and P_k is R_k with its columns shifted one step to the right. We shall prove

$$\det M_n = \det M_{\lfloor n/2\rfloor} \cdot \det R_{\lceil n/2\rceil}, \qquad \det R_k = (-1)^{\lfloor k/2\rfloor}, \qquad \det P_k = \det M_k. \tag{1}$$

Lemma (One binary digit at a time)**.** *Let $a = 2a' + \beta$ and $c = 2c' + \delta$ with $\beta, \delta \in \{0,1\}$. Then*

$$G(a,c) = \begin{cases} G(a',c') & \text{if } \beta = 0, \\ \delta \cdot G(a',c'+1) & \text{if } \beta = 1. \end{cases}$$

Proof. The lowest binary digit of $a + c$ is $\beta + \delta \bmod 2$, the carry into the next position is $\lfloor(\beta+\delta)/2\rfloor$, and the remaining digits of $a + c$ are those of $a' + c' + \lfloor(\beta+\delta)/2\rfloor$. If $\beta = 0$, the lowest digit of a is 0, so it imposes no condition, there is no carry, and $a \,\&\, (a + c) = 0$ reduces to $a' \,\&\, (a' + c') = 0$. If $\beta = 1$, the lowest digit of $a + c$ must be 0, which forces $\delta = 1$; then the carry is 1 and the condition becomes $a' \,\&\, (a' + c' + 1) = 0$. □

Lemma (Reduction of M_n and of P_k)**.** *For $n \ge 1$, $\det M_n = \det M_{\lfloor n/2\rfloor} \cdot \det R_{\lceil n/2\rceil}$, and for $k \ge 1$, $\det P_k = \det P_{\lfloor k/2\rfloor} \cdot \det R_{\lceil k/2\rceil}$. Consequently $\det P_k = \det M_k$ for all $k \ge 0$.*

Proof. Take M_n first. For every $c' \ge 1$ with $2c' + 1 \le n$, subtract column $2c'$ from column $2c' + 1$; this does not change the determinant. Consider a row of even index $a = 2a'$ with $a' \ge 1$. By the previous lemma its entries in columns $2c'$ and $2c' + 1$ are both $G(a', c')$, so after the subtraction it has 0 in every odd column $c \ge 3$; and in column 1 it has $G(a', 0) = 0$. A row of odd index $a = 2a' + 1$ has 0 in every even column (the case $\delta = 0$), is untouched by the subtractions, and in the odd column $c = 2c' + 1$ carries $G(a', c' + 1)$. Now list the even indices first and the odd indices second, for the rows and for the columns alike; since the same permutation is applied to rows and to columns, the determinant is unchanged. The result is block diagonal: the even rows against the even columns give $\big(G(a',c')\big)_{1\le a',c'\le\lfloor n/2\rfloor} = M_{\lfloor n/2\rfloor}$, and the odd rows against the odd columns give $\big(G(a',c'+1)\big)_{0\le a',c'\le\lceil n/2\rceil-1} = R_{\lceil n/2\rceil}$.

For P_k the rows are $a = 0, \dots, k-1$ and the columns are $c = 2, \dots, k+1$. Subtract column $2c'$ from column $2c' + 1$ whenever both lie in this range. The even rows $a = 2a'$ ($0 \le a' < \lceil k/2\rceil$) are then 0 in the odd columns and carry $G(a', c')$ in the even columns $c = 2c'$, $1 \le c' \le \lceil k/2\rceil$; the odd rows $a = 2a' + 1$ ($0 \le a' < \lfloor k/2\rfloor$) are 0 in the even columns and carry $G(a', c' + 1)$ in the odd columns $c = 2c' + 1$, $1 \le c' \le \lfloor k/2\rfloor$. Column c sits in position $c - 2$, of the same parity as c, so again one permutation serves rows and columns, and the matrix becomes block diagonal with blocks $R_{\lceil k/2\rceil}$ and $\big(G(a',c'')\big)_{0\le a'<\lfloor k/2\rfloor,\ 2\le c''\le\lfloor k/2\rfloor+1} = P_{\lfloor k/2\rfloor}$.

The two sequences $\det M_k$ and $\det P_k$ satisfy the same recursion with the same initial values $\det M_0 = \det P_0 = 1$, $\det M_1 = \det P_1 = 1$ (indeed $M_1 = (G(1,1)) = (1)$ and $P_1 = (G(0,2)) = (1)$), so they coincide. □

Lemma (The companion matrix)**.** *For every $k \ge 1$, $\det R_k = (-1)^{\lfloor k/2\rfloor}$.*

Proof. We argue by strong induction on k; the case $k = 1$ is $R_1 = (G(0,1)) = (1)$. The inductive hypothesis, with the previous lemma, gives $\det R_m = \pm 1$ and $\det M_m = \det P_m = \pm 1$ for all $m < k$. The rows of R_k are $a = 0, \dots, k-1$, the columns $c = 1, \dots, k$. Subtract column $2c'$ from column $2c'+1$ for every $c' \geq 1$ with $2c'+1 \leq k$. Afterwards:

- an even row $a = 2a'$ has entry $G(a',0) = [a' = 0]$ in column 1, entry $G(a',c')$ in the even column $2c'$, and 0 in every odd column $c \geq 3$;
- an odd row $a = 2a'+1$ has entry $G(a',1)$ in column 1, entry $G(a',c'+1)$ in the odd column $2c'+1$ with $c' \geq 1$, and 0 in every even column.

Column 1 is the one column shared by both kinds of rows.

Case $k = 2q$. There are q even rows ($0 \leq a' < q$), q odd rows ($0 \leq a' < q$), the q even columns $2, 4, \dots, 2q$, and the $q-1$ odd columns $3, 5, \dots, 2q-1$. Order the rows as (even, odd) and the columns as (even columns, column 1, odd columns). The matrix takes the form

$$\begin{pmatrix} E & e_0 & 0 \\ 0 & v & F \end{pmatrix}, \qquad E = \big(G(a',c')\big)_{0 \leq a' < q,\ 1 \leq c' \leq q} = R_q,$$

$$v = \big(G(a',1)\big)_{0 \leq a' < q}, \qquad F = \big(G(a',c'')\big)_{0 \leq a' < q,\ 2 \leq c'' \leq q},$$

where e_0 is the column with a 1 in the first place and 0 elsewhere. The block $(v \mid F)$ is exactly R_q again. So the matrix is block upper triangular with diagonal blocks R_q and R_q, and its determinant is $(\det R_q)^2 = 1$. It remains to count inversions of the two reorderings. Rows: the order $(0, 2, \dots, 2q-2, 1, 3, \dots, 2q-1)$ has $\binom{q}{2}$ inversions. Columns: column c originally sits in position $c-1$, so the new order is (positions $1, 3, \dots, 2q-1$, then 0, then $2, 4, \dots, 2q-2$), with $q + \binom{q}{2}$ inversions. The total, q^2, has the parity of q, so $\det R_{2q} = (-1)^q$.

Case $k = 2q+1$. Now there are $q+1$ even rows ($0 \leq a' \leq q$), q odd rows, the q even columns $2, \dots, 2q$ and the q odd columns $3, \dots, 2q+1$. Order the rows as (even, odd) and the columns as (column 1, even columns, odd columns):

$$\begin{pmatrix} e_0 & E' & 0 \\ v & 0 & F' \end{pmatrix}, \qquad E' = \big(G(a',c')\big)_{0 \leq a' \leq q,\ 1 \leq c' \leq q}, \quad F' = \big(G(a',c'')\big)_{0 \leq a' < q,\ 2 \leq c'' \leq q+1} = P_q.$$

Expand along the first column. Its entry in the even row $a' = 0$ is 1 and, deleting that row and the column, what remains is block diagonal with blocks $\big(G(a',c')\big)_{1 \leq a', c' \leq q} = M_q$ and $F' = P_q$, so this term equals $\det M_q \det P_q = (\det M_q)^2 = 1$. Deleting instead an odd row and the first column leaves the block matrix $\left(\begin{smallmatrix} E' & 0 \\ 0 & F'' \end{smallmatrix}\right)$ with E' of size $(q+1) \times q$ and F'' of size $(q-1) \times q$; its rank is at most $q + (q-1) < 2q$, so its determinant is 0. Hence the reordered matrix has determinant 1. Inversions: rows $(0, 2, \dots, 2q, 1, 3, \dots, 2q-1)$ give $\binom{q+1}{2}$; columns, in positions $(0, 1, 3, \dots, 2q-1, 2, 4, \dots, 2q)$, give $\binom{q}{2}$. The total is again q^2, so $\det R_{2q+1} = (-1)^q$. □

Conclusion. Write $n_t = \lfloor n/2^t \rfloor$. Iterating the first relation in (1) along $n_0 = n$, n_1, $n_2, \dots$ until the index reaches 0, and inserting $\det R_k = (-1)^{\lfloor k/2 \rfloor}$, gives

$$\det M_n = (-1)^{E(n)}, \qquad E(n) = \sum_{t \geq 0} \left\lfloor \tfrac{1}{2} \lceil n_t/2 \rceil \right\rfloor = \sum_{t \geq 0} \left\lfloor \frac{n_t + 1}{4} \right\rfloor = \sum_{t \geq 0} \left\lfloor \frac{n + 2^t}{2^{t+2}} \right\rfloor,$$

using $\lfloor \lceil m/2 \rceil / 2 \rfloor = \lfloor (m+1)/4 \rfloor$ and $\lfloor (\lfloor x \rfloor + 1)/4 \rfloor = \lfloor (x+1)/4 \rfloor$. Fix t and write $n = 2^{t+2}Q + r$ with $0 \leq r < 2^{t+2}$. Then $(n + 2^t)/2^{t+2} = Q + (r + 2^t)/2^{t+2}$ and $0 < (r + 2^t)/2^{t+2} < 2$, so the floor is $Q+1$ when $r \geq 3 \cdot 2^t$ and Q otherwise; and $r \geq 3 \cdot 2^t$ means precisely that the binary digits of n in positions t and $t+1$ are both 1. Therefore

$$E(n) = \sum_{u \geq 2} \left\lfloor \frac{n}{2^u} \right\rfloor + p(n),$$

where $p(n)$ is the number of positions t such that digits t and $t+1$ of n are both 1. If $s(n)$ is the number of digits 1 of n, then $\sum_{u\ge1}\lfloor n/2^u\rfloor = n - s(n)$ (each digit 1 in position ℓ contributes $2^{\ell-1}+\cdots+1 = 2^\ell - 1$), so

$$E(n) = n - s(n) - \left\lfloor \frac{n}{2} \right\rfloor + p(n) = \left\lceil \frac{n}{2} \right\rceil - \big(s(n) - p(n)\big).$$

Finally a block of ℓ consecutive 1's contributes ℓ to $s(n)$ and $\ell - 1$ to $p(n)$, so $s(n) - p(n) = b$, the number of blocks. Hence $E(n) \equiv \lceil n/2 \rceil + b \pmod 2$ and $\det M = (-1)^{\lceil n/2\rceil + b}$. □

Remarks. Since $\sum_{t\ge0}\lfloor (n_t+1)/4 \rfloor$ counts the integers $k \le n$ whose odd part is 3 mod 4, the ratio $\det M_n / \det M_{n-1}$ is -1 exactly when the odd part of n is 3 mod 4: this is the regular paperfolding sequence, so $\det M_n$ is also the product of the first n paperfolding signs. For $n = 2026 = 11111101010_2$ one has $b = 3$ and $\lceil n/2 \rceil = 1013$, so $\det M = 1$; for $n = 2027$ the blocks are 111111, 1, 11, again $b = 3$, and $\det M = -1$. The proof shows more: every leading principal minor of M is ±1, so M is unimodular and so is its inverse.

Why it is hard, and why it is original

Why it is hard. The parity of $\binom{2i+j}{i}$ looks like an invitation to Lucas's theorem, and indeed the first step is routine. The trap is the natural next step. The matrix $\big(\binom{i+j}{i} \bmod 2\big)_{0\le i,j<n}$, the familiar Sierpinski pattern, factors digit by digit as a tensor power of $\left(\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right)$, and any matrix of the form $f(i \,\&\, j)$ on an initial segment of the integers factors as $ZDZ^{\mathsf T}$ with Z the unitriangular subset-inclusion matrix, so its determinant is a product of Möbius transforms, a direct cousin of Smith's determinant. Here the column index is sheared by the row index: the entry is $\binom{i+(i+j)}{i} \bmod 2$, so the carries in $i + j$ couple the binary digits and no digit-by-digit factorisation exists. The parity split survives, but it hands back a matrix with an extra all-ones row, and that row cannot be removed; the induction only closes after a second companion family is introduced and the shared column is understood. Even with ±1 in hand, the sign is the hard part: it is a product of paperfolding signs, something no small-case computation suggests, and it falls out of inversion counts of two different interleavings that happen to agree modulo 2.

What makes it original. The determinants of the Sierpinski matrix itself and of its border submatrices are classical: R. Bacher (*Matrices autosimilaires*, 2002) obtained the Thue–Morse sign for $\det\big(\binom{i+j}{i} \bmod 2\big)_{0\le i,j<n}$, and M. Mereb (2022) proved M. Levin's assertion that every square submatrix on the upper or left border of that matrix has determinant ±1. The present matrix is a diagonal shear of the same pattern, which lies outside both of those self-similar classes, and its sign statistic, the number of blocks of consecutive 1's, is different in kind from the digit sum that governs the unsheared case. The nearest relatives in spirit are the Hankel determinants of the paperfolding and Thue–Morse sequences (Allouche, Peyrière, Wen and Wen), where the same automatic sequences appear, though for a different reason.

Verification

Exact integer determinants (fraction-free elimination) of M_n for every $n \le 180$ agree with $(-1)^{\lceil n/2\rceil+b}$ in all 180 cases, and the matrix built from $\binom{2i+j}{i} \bmod 2$ was checked entry by entry against the bitwise description for the same range. The determinants of M_{2026} and M_{2027} were computed modulo the prime $2^{31} - 1$ by Gaussian elimination: $+1$ and -1, as the formula predicts; a floating-point `slogdet` gave the same signs. The companion families were checked directly: $\det R_k = (-1)^{\lfloor k/2\rfloor}$ and $\det P_k = \det M_k$ for all $k \le 70$, together with the recursion $\det M_n = \det M_{\lfloor n/2\rfloor} \det R_{\lceil n/2\rceil}$ for $n \le 70$. The three closed forms (ceiling plus block count; $(n - r)/2$ with r the number of runs of equal digits; the paperfolding count of $k \le n$ with odd part 3 mod 4) were compared for all $n \le 10^5$ and agree.